

Summary

PhD Candidate in Mechanical Engineering at Purdue University with 3+ years developing perception systems for autonomous vehicles. Expertise in building real-time perception pipelines, multi-modal sensor fusion architectures, and deep learning models deployed on GPU-accelerated platforms including NVIDIA Jetson. Experienced in scaling perception models from research to deployment, designing evaluation frameworks, and collaborating with cross-functional autonomy teams to deliver robust AV software modules for real-world environments.

Education

Purdue University Doctor of Philosophy in Mechanical Engineering <i>Thesis: Hierarchical Radar-Camera fusion for task-sufficient 3D object and obstacle zone detection in autonomous navigation using minimal sensor information</i>	Indianapolis, IN December 2026 (expected)
Purdue University Master of Science in Mechanical Engineering GPA: 3.94/4.00	Indianapolis, IN December 2024
Bangladesh University of Engineering and Technology Bachelor of Science in Mechanical Engineering GPA: 3.76/4.00 <i>Dean's Award, Technical Scholarship, University Merit Scholarship</i>	Dhaka, Bangladesh August 2017

Skills

Perception & Computer Vision: Object detection, multi-object tracking, semantic segmentation, 3D perception, BEV encoding
Sensor Fusion & Localization: Multimodal fusion (camera, radar, LiDAR), Kalman filtering, SLAM, visual odometry, state estimation, data association, time synchronization
Machine Learning: PyTorch, TensorFlow, CNNs, attention transformers, model optimization, ablation studies
Autonomous Systems: ROS/ROS2, real-time perception pipelines, GPU acceleration (CUDA), embedded deployment (NVIDIA Jetson), motion planning, path planning
Programming & Tools: Python, C++, MATLAB, OpenCV, NumPy, Pandas, point cloud processing, Unreal Engine
Evaluation & Testing: Benchmark development, error analysis, edge case identification, hard negative mining, performance metrics (MOTA, mAP, IoU)

Research Experience

<i>Research Assistant, Mechatronics and Autonomous Research Lab (MARL), Purdue University</i> Indianapolis, IN	August 2022 - Present
Off-Road Semantic Segmentation Dataset and Deep Learning Model: - Built a dataset pipeline of 1,414 images through label refinement of Yamaha-CMU Off-Road Dataset and class-balancing via synthetic image augmentation to train real-time perception models. - Designed ResNet34-U-Net semantic segmentation architecture achieving 87% accuracy at 24 Hz demonstrating strong cross-dataset generalization on YCOR, Rellis-3D, and RUGD benchmarks without fine-tuning.	
PrED: Predictive Enhancement of Detection Algorithm - Designed a novel object detection and enhancement algorithm with Kalman Filter and Predictability Index to recover low-confidence and missed detections in data-limited scenarios. - Implemented CUDA-accelerated inference, reducing detection latency from 500ms to 79ms and enabling real-time object detection for autonomous navigation. - Improved detection reliability with +11% detection accuracy and +7% MOTA score in moving camera off-road experiments.	
Semantic-guided Hierarchical Multi-modal sensor fusion for end-to-end perception pipeline: - Developing a three-stage radar-camera fusion with semantic-guided proposal generation to achieve task-sufficient 3D object and obstacle zone detection using only minimal sensor information. - Modified YOLOv8 architecture with radar and semantic BEV encoders for 3D bounding box prediction. - Integrated the PrED algorithm into 3D detection pipeline, improving temporal detection stability in dynamic environments. - Lifted terrain semantics from segmentation outputs to BEV plane for terrain traversability estimation in off-road autonomous navigation. - Trained models on 5k+ annotated frames and radar sweeps of NuScenes dataset. - Integrating TensorRT-optimized inference pipelines, reducing model latency and enabling real-time deployment on NVIDIA Jetson platforms. - Designing the architecture as a complete perception pipeline supporting navigation and motion planning modules, generating obstacle zones and traversability maps for downstream path planning.	
Physics-based Radar Sensor Model in Unreal Engine: - Built high-fidelity automotive radar sensor model in Unreal Engine C++ integrating physics-based signal characteristics, point cloud processing, and 3D grid-DBSCAN clustering. - Achieved 89% correlation with real-world automotive radar measurements, enabling sim-to-real validation of perception and control algorithms.	
Cross-Functional Collaboration: - Collaborated with 10+ multidisciplinary team members on DoD-funded autonomous vehicle projects, delivering production-ready autonomy stack features such as ROS2 module development through systematic evaluation and iterative improvement. - Mentored teams developing edge-AI models deployed on NVIDIA Jetson Orin Nano platforms.	

Industry Experience

<i>CM Digital Transformation Lead and Assistant Quality Manager, Unilever</i> Dhaka, Bangladesh	April 2018 - July 2022
- Led industrial automation initiatives across 3 manufacturing plants, implementing sensor-based monitoring systems and real-time analytics. - Developed data-driven production monitoring systems, reducing operational inefficiencies by 40%, receiving 2× Supply Chain Director's Awards. - Implemented predictive maintenance pipelines using real-time sensor analytics. - Managed cross-functional teams integrating hardware sensors, data pipelines, and production analytics systems.	

Publications

Rahi, A., Wasfy, H., Wasfy, T., & Anwar, S., <i>Improved Object Detection and Tracking with Camera in Motion Using PrED: Predictive Enhancement of Detection</i>, Automation, 6(4), (2025), 77. DOI: 10.3390/automation6040077	
---	--

- Khan, M. N., **Rahi, A.**, Hasan, M. A., & Anwar, S., *Decision-Level Multi-Sensor Fusion to Improve Limitations of Single-Camera-Based CNN Classification in Precision Farming: Application in Weed Detection*, *Computation*, 13(7), (2025), 174. DOI: [10.3390/computation13070174](https://doi.org/10.3390/computation13070174)
- **Rahi, A.**, Cardoza, C. O., Vemulapalli, S. S. T., El-Mounayri, H., et al., *Modeling of an Automotive Radar Utilizing Grid-DBSCAN and SNR Characteristics of Virtual Objects in Unreal Engine*, *IEEE Access*, 12, (2024), 127836-127845. DOI: [10.1109/ACCESS.2024.3455189](https://doi.org/10.1109/ACCESS.2024.3455189)
- **Rahi, A.**, Elgeoushy, O., Syed, S. H., El-Mounayri, H., et al., *Deep Semantic Segmentation for Identifying Traversable Terrain in Off-Road Autonomous Driving*, *IEEE Access*, 12, (2024), 162977-162989. DOI: [10.1109/ACCESS.2024.3491135](https://doi.org/10.1109/ACCESS.2024.3491135)
- Khan, M. N., **Rahi, A.**, Rajendran, V. P., Hasan, M. A., & Anwar, S., *Real-time Crop Row Detection Using Computer Vision - Application in Agricultural Robots*, *Frontiers in Artificial Intelligence*, 7, (2024), 1435686. DOI: [10.3389/frac.2024.1435686](https://doi.org/10.3389/frac.2024.1435686)
- Parkar, O., Snyder, B., **Rahi, A.**, & Anwar, S., *Modified Particle Swarm Optimization Based Powertrain Energy Management for Range Extended Electric Vehicle*, *Energies*, 16(13), (2023), 5082. DOI: [10.3390/en16135082](https://doi.org/10.3390/en16135082)

Conference Proceedings

- **Rahi, A.**, Cardoza, C. O., Vemulapalli, S. S. T., Wasfy, T., & Anwar, S., *Modeling of Automotive Radar Sensor in Unreal Engine for Autonomous Vehicle Simulation*, ASME International Mechanical Engineering Congress and Exposition, (2023), 87639. DOI: [10.1115/IMECE2023-112964](https://doi.org/10.1115/IMECE2023-112964)
- Vemulapalli, S. S. T., **Rahi, A.**, Mullins, S. D., Wasfy, H., et al., *IR Sensor Modeling in Unreal Engine for Autonomous Vehicle Applications*, ASME International Mechanical Engineering Congress and Exposition, (2023), 87639. DOI: [10.1115/IMECE2023-113001](https://doi.org/10.1115/IMECE2023-113001)
- Barua, S., **Rahi, A.**, Niloy, R. A., & Tahmid, S., *Numerical Analysis of Shock Oscillation Around a Biconvex Circular Arc Airfoil by Incorporating a Bump on It in a Channel*, International Conference on Mechanical Industrial and Materials Engineering, (2017).
- Mahmud, Z., Azad, R. I., Niloy, R. A., **Rahi, A.**, Tahmid, S., *Observation of Unsteady Shock Behavior around a Biconvex Circular Arc Airfoil with Circular Cavities on it in a channel*, International Conference on Mechanical Industrial and Materials Engineering, (2017).
- Fattah, S. A., Haider, M. Z., Chowdhury, D., Sarkar, M., Chowdhury, R. I., Islam, M. S., ... **Rahi, A.**, *An Aerial Landmine Detection System with Dynamic Path and Explosion Mode Identification Features*, 2016 IEEE Global Humanitarian Technology Conference (GHTC), (2016), 745-752.